

PROJECT DOCUMENTATION PORTFOLIO

Meta Ad Performance Analysis

A Business Intelligence Case Study on Facebook & Instagram Paid Campaign Performance

PLATFORMS COVERED

Facebook · Instagram

ANALYSIS WINDOW

June 2025

REPORTING TOOL

Power BI

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PART ONE

Business Requirements

Why this report exists, what it covers, and how success is measured.

The Business Need

The marketing organization runs paid campaigns across Facebook and Instagram but has no unified way to see how those campaigns are performing. This report was commissioned to close that gap by giving the marketing team one place to see reach, engagement, conversions, and budget efficiency across both platforms.

With that visibility, the team can do three things it couldn't do reliably before:

- Identify which platform (Facebook or Instagram) is the stronger performer, and why.
- Track return on ad spends and reallocate budget toward what's working.
- Understand audience engagement patterns well enough to target more precisely.

Scope

SCOPE	DETAIL
In scope	Paid campaigns running on Facebook and Instagram.
Out of scope	Other Meta placements (Messenger, Audience Network).
Out of scope	Organic engagement, only paid ad activity is measured.

KPI Glossary

Every number on the dashboard traces back to one of these definitions — nothing is calculated ad hoc.

KPI	DEFINITION	FORMULA	USED FOR
Impressions	Times an ad was displayed	COUNT(event_type = Impression)	Measuring reach
Clicks	Times a user clicked an ad	COUNT(event_type = Click)	Engagement intent
Shares	Times an ad was shared	COUNT(event_type = Share)	Viral engagement
Comments	User comments left on an ad	COUNT(event_type = Comment)	Sentiment & feedback

KPI	DEFINITION	FORMULA	USED FOR
Purchases	Purchases made after seeing an ad	COUNT(event_type = Purchase)	Conversions
Engagements	Total interaction volume	Clicks + Shares + Comments	Engagement volume
CTR	% of impressions that became clicks	(Clicks ÷ Impressions) × 100	Ad effectiveness
Engagement Rate	% of impressions that became engagements	(Engagements ÷ Impressions) × 100	Overall ad appeal
Conversion Rate	% of clicks that became purchases	(Purchases ÷ Clicks) × 100	Funnel efficiency
Purchase Rate	% of impressions that became purchases	(Purchases ÷ Impressions) × 100	Reach-to-sale efficiency
Total Budget	Total ad spend across campaigns	SUM(campaigns.total_budget)	Cost analysis
Avg. Budget / Campaign	Average spend per campaign	Total Budget ÷ Campaign Count	Budget distribution

Required Visuals

Seven visuals were specified, each answering one business question and all re-slicing dynamically by whichever KPI is selected.

#	VISUAL	TYPE	BUSINESS QUESTION
1	Target Gender	Donut chart	Which gender segment drives the most value?
2	Target Age Group	Bar chart	Which age bracket is most responsive?
3	Country	Map	Where is performance strongest geographically?
4	Calendar Month	Heat map	When do campaigns peak — seasonally or by date?
5	Weekly Trend by Ad Type	Stacked column	How do formats trend week over week?
6	Hourly Trend	Area chart	What time of day performs best?
7	Ad Type × Platform	Matrix	How do formats compare across Facebook and Instagram?

PART TWO

Data Architecture

How the underlying dataset is structured, and why it's modeled this way.

About the Dataset

The data mirrors how Meta's ad platforms capture information: campaign-level strategy, ad-level creative and targeting metadata, user-level demographics, and an event-level log of every interaction. Together, they support the full set of KPIs above including cost-efficiency metrics like CPC, CPM, and ROAS and are built specifically to power a Power BI dashboard evaluating campaign effectiveness, budget use, and engagement patterns.

Star Schema

One fact table sits at the center, surrounded by three-dimension tables:

FACT TABLE

ad_events — every impression, click, share, comment, and purchase. Every KPI in this report is derived from this table.

- ad_events → ads — links each event to the ad's platform and creative type.
- ads → campaigns — links ads to the campaign's budget and timeframe.
- ad_events → users — links each event to the demographics of the person behind it.

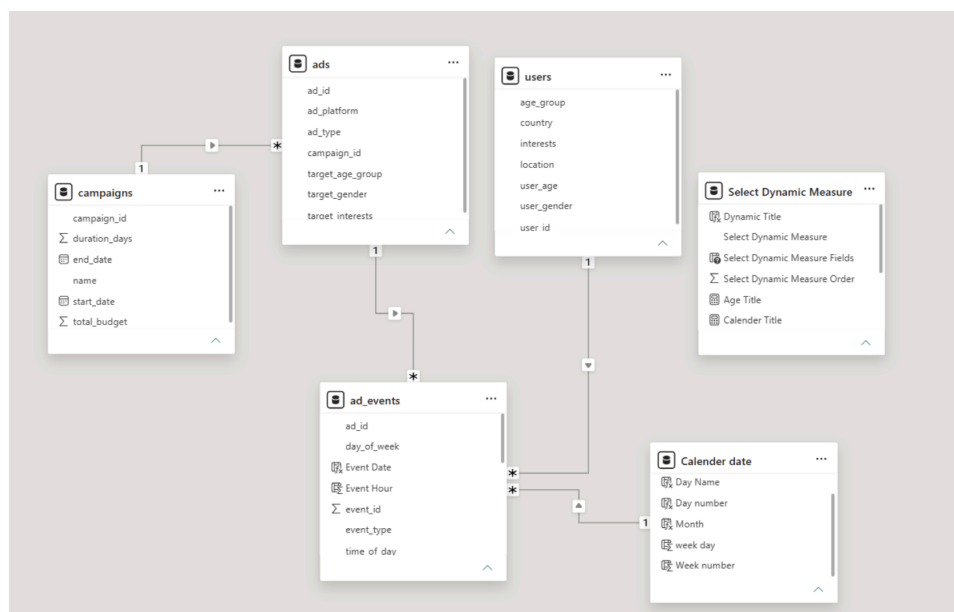


Figure 1 — The Power BI data model: one fact table, four dimensions, and a Calendar table driving all time-based visuals.

Table Reference

AD_EVENTS — EVENT LOG (FACT TABLE)

Captures every interaction between a user and an ad; the foundation for Impressions, Clicks, CTR, Conversion Rate, and ROAS.

FIELD	DESCRIPTION	EXAMPLE
event_id	Unique event identifier — primary key	100234
ad_id	Links to the ads table	501
user_id	Links to the users table	U_1204
timestamp	Exact time of the event	2025-03-12 14:30:00
day_of_week	Derived: day of the week	Tuesday
time_of_day	Derived: segment of the day	Afternoon
event_type	Impression, Click, Share, Comment, or Purchase	Click

ADS — CREATIVE & TARGETING

Defines each ad's metadata — which platform it ran on, its format, and who it targeted.

FIELD	DESCRIPTION	EXAMPLE
ad_id	Unique ad identifier — primary key	501
campaign_id	Links to the campaigns table	C_101
ad_platform	Facebook, Instagram, Messenger, or Audience Network	Instagram
ad_type	Image, Video, Carousel, or Story	Video
target_gender	Gender the ad was targeted at	Female
target_age_group	Age bracket the ad was targeted at	25–34
target_interests	Topics/interests targeted	Travel, Fashion

CAMPAIGNS — BUDGET & STRATEGY

Campaign-level context: what it's called, how long it ran, and what it spent.

FIELD	DESCRIPTION	EXAMPLE
campaign_id	Unique campaign identifier — primary key	C_101
name	Campaign name	“Spring Promo 2025”
start_date / end_date	Campaign window	2025-03-01 → 2025-03-31
duration_days	Derived: campaign length	30
total_budget	Budget allocated to the campaign	\$50,000

USERS — AUDIENCE DEMOGRAPHICS

Who engaged: demographics and interests, used for segmentation and targeting-accuracy checks.

FIELD	DESCRIPTION	EXAMPLE
user_id	Unique user identifier — primary key	U_1204
user_gender / user_age	Demographics of the user	Male, 27
age_group	Grouped age bucket	25–34
country / location	Geography, down to city/state	India / Bangalore
interests	User's stated interests	Tech, Travel

PART THREE

Findings & Insights

What a full month of paid Facebook and Instagram activity actually looked like.

The Headline Numbers

METRIC	RESULT	READ
Impressions	216K	Solid reach
Clicks	25.4K	—
CTR	11.76%	Well above the ~1–2% industry norm
Engagement Rate	13.56%	Audience is genuinely interested
Purchases	1.3K	—
Conversion Rate (click → purchase)	5.21%	Reasonable, but improvable
Purchase Rate (impression → purchase)	0.61%	The real bottleneck
Total Budget	\$2.5M across ~50.7K/campaign	—

THE STORY HERE

Creative and targeting are clearly working, people are seeing the ads and reacting to them. The drop happens after the click. This is a funnel with a strong top and a weak bottom.

Who's Engaging

Engagement skews young and female. Women account for 43% of engagement versus 22% for men (the remaining 35% unspecified), nearly a 2:1 gap. By age, activity peaks in the 20–30 bracket, especially the early twenties, and falls off sharply past 35. The effective audience for this campaign set is young adult women, whether targeting was explicitly built around that.

Where It's Happening

The US, India, Brazil, Germany, and the UK are top engagement contributors, with India and Brazil providing volume and Germany and the UK higher purchasing power.

When It's Happening

It's interesting to see that engagement has been consistent each week, with no signs of weariness or sudden drops. However, there's a noticeable pattern: activity tends to be at its highest between 3:00 PM and 8:00 PM, and it's lowest overnight (12:00 AM to 5:00 AM). Interestingly, engagement also increased on certain days throughout the month, specifically the 19th–21st and 25th–27th, which seems to coincide with promotional or launch events rather than just random fluctuations.

What's Working — By Ad Format

AD TYPE	IMPRESSIONS	CLICKS	CTR	PURCHASE RATE	CONVERSION RATE	ENGAGEMENT RATE
Video	46K	5K	11.9%	0.62%	5.2%	13.7%
Stories	72K	8K	11.8%	0.65%	5.2%	13.6%
Carousel	48K	6K	11.7%	0.59%	5.1%	13.4%
Image	51K	6K	11.7%	0.57%	4.9%	13.5%

Video wins on CTR, conversion rate, and engagement rate per impression. Stories win on sheer reach and edges out on purchase rate. Image and Carousel are the consistent laggards across every efficiency metric which is not broken, just outperformed.

The Dashboard, Side by Side

Facebook and Instagram each get a dedicated page in the report, built on the same model, so results are genuinely comparable rather than eyeballed across two separate tools.

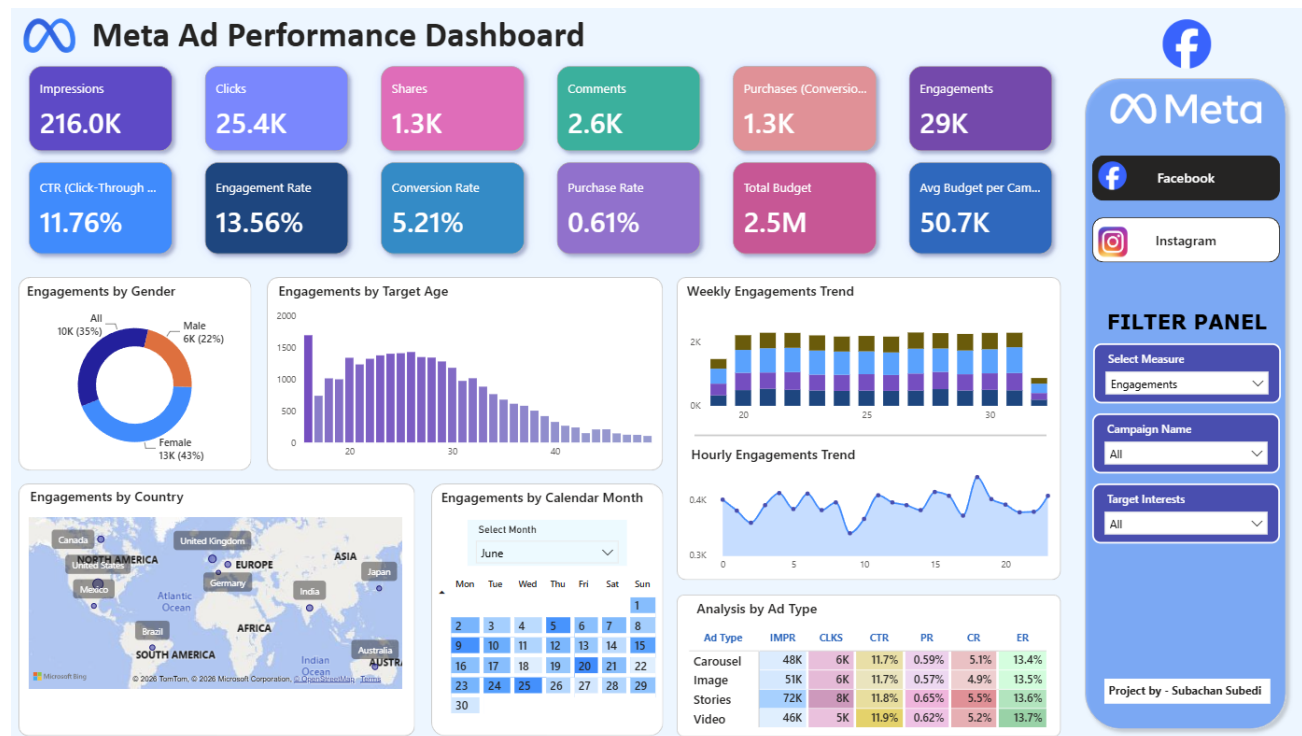


Figure 2 — Facebook performance view.

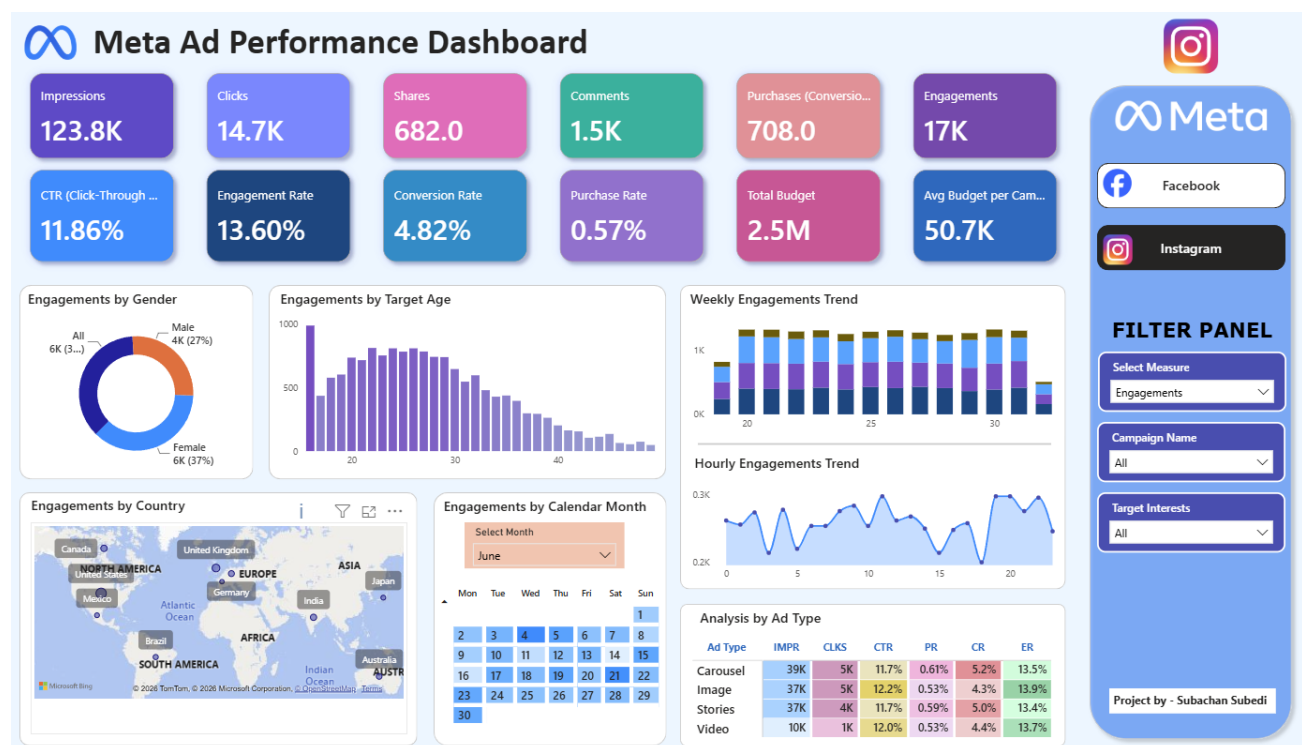


Figure 3 — Instagram performance view.

PART FOUR

Recommendations

What the data suggests should happen next.

1. Fix the leak, not the funnel top. CTR and engagement are already strong — the real ROI lever is landing page experience, offer relevance, and retargeting, not more top-of-funnel spend.
2. Concentrate targeting on females aged 18–30. The segment already driving disproportionate engagement.
3. Shift budget toward Video and Stories, and away from Image/Carousel, based on measured efficiency rather than assumption.
4. Schedule ad delivery for the afternoon–evening window (roughly 15:00–20:00), where engagement consistently peaks.
5. Differentiate by geography. Scale spends in India and Brazil for volume; invest in creative and offer quality in Germany and the UK to capture higher-value conversions.
6. Lean into promotion-driven spikes. The calendar pattern suggests scheduled campaigns and launches reliably outperform steady state spend.